

Unit 4

Medicinal Substance - Dravya (Drug)

Visited -
Iskon
Varanasi

Date _____ Page _____

- Dravya - A substance which possesses Guna (property) and Karma (action) with inheritance relation.
- five factors - Rasa, Virya, Vipaka, guna prabhava depend on dravyas.
- The character of Dravya is the guna & Karma is on it with inheritance relation (Samavaya sambandha)
- Rasapanchaka - five main attributes of dravya

- (i) Rasa - The taste of the substance.
- (ii) Guna - The qualities or basic nature of the substance.
- (iii) Veerya - The potency of the substance.
- (iv) Vipaka - The taste conversion after digestion or bio-transformation.
- (v) Prabhava - The special effect of the substance.

• Strength $\rightarrow$ (i) Prabhava $>$ (ii) Virya $>$ (iii) Vipaka $>$ (iv) Rasa

➤ Importance of Dravya (Drug)

- (i) Vyavasthitatva - Stability.
 - $\rightarrow$ Dravya is stable in nature.
 - $\rightarrow$ But Rasa, Guna, Vipaka, Virya - unstable
- (ii) Ashrayatva - The Dravya can host for the component - Rasa, Guna, Virya etc.
- (iii) Abhasamarthya - The initiation of specific pharmacologically action is possible only through using Dravya.
- (iv) Ekalesha Sadhyatva - Dravya can be used treatment according to the grading like.
- (v) Karma Apekshitatva - The properties depend upon Dravya.

° Dravya Vargikarana - classification of Drugs

Date _____ Page _____

(1) Karyā Dravyas - which are anitya (~~infinte~~)

(i) Achetana Karyā Dravyas - The Dravyas with Indriya or life ex - Humans, Plants, Animals.

* Sthavara (Antaschetana) - life present inherently but cannot be expressed out. eg - plants.

* Jangama (Bahir Antaschetana) - The life can be exhibited out. eg - Human, ~~plants~~ Animals.

° Bahir Antaschetana is divided into 4 types.

(a) Jarayuja - Rāsa Placental origin - Human, animals - lion

(b) Andaja - Egg origin - Birds, Serpents, fish etc.

(iii) Swedaja - Sweat origin - Born from the mud or clay.
eg - Frogs, Indragopa Insects.

(c) Udbhija - Soil origin - Born from the mud or clay
eg - Frogs, Indragopa Insects.

(d) Swedaja - Sweat origin - The creatures which have their birth in Sweda or dirty water. eg - Insects, mosquito.

(ii) Achetana Karyā Dravya - The Dravyas without Indriya, Nijiva or without life force eg - Swarna, Rajata etc. Mukta etc.

(2) Karana Dravyas - which are Nitya.

→ 9 in Number - Akasha, Vayu, Agni, Jala, Pithwi, Atma, Manas, Kala, Dik.

→ Responsible for the production of innumerable Karyā Dravyas.

➤ Classification of dravyas on the basis of Origin.

→ three types

Date _____ Page _____

- ① Jangama Dravyas - The Dravyas of animal origin.
→ Jaryuja, Andaja, Swedaja, Udbhija types of Jangama.
- ② Bhauma or Paathiva Dravyas - ^{beneath}
The Dravyas obtained from the ~~above~~ the earth ex-
minerals, and metals.
- ③ Audbhida Dravyas - The Dravyas obtained from
above the earth, ex- Plants
→ Also four types - (Anatasch-efana Dravya)
 - (i) Vanaspathi - hidden flower, visible fruit
 - (ii) Vanaspathya - Have both flowers and fruits
 - (iii) Aushadi - Plants spread on the ground with branches
and die after producing fruit
 - (iv) Virudha - Plants that vanish after their fruit mature.

Medicinal Plants

Date _____ Page _____

- Nature itself is the best Remedy.
- Mother nature offers medicine for many diseases through plants and herbs.

➤ A 'medicinal plant' is one that possesses medicinal properties that can treat or cure ailments.

➤ Top medicinal plants given below -

① Tulsi - "queen of Herbs"

- leaves - are rich in antioxidants & anti-fungal properties.
- Antioxidants - regulate the blood glucose level and boost the immune systems.
- Anti-fungal - best cure for cold and flu.
- It eases digestive problems and promotes overall health.

② Aloe-Vera - have antioxidant and antimicrobial properties help to ease constipation, detoxify the body and help to treat wounds.

- Aloe vera is well known for its abundant beauty benefits.
- It helps to treat wrinkles, moisturises dry skin, treat Eczema, promotes hair growth. Commercial - hair gel, face-wash.

③ Fenugreek (Methi) - have therapeutic Properties.

- fenugreek Seed used ingredient.
- Seed have galactagogue properties - increase breast milk production.
 - ↳ also relieve PMS and menopause symptoms.
 - ↳ In Man seeds help to Regulate the Testosterone hormone
- Seeds reduce cholesterol levels, lower blood glucose level.
- also promotes hair growth and aids weight loss.

④ Ashwagandha - All plants part (roots, leaves, fruits) exhibit medicinal properties.

Help to treat - Insomnia, reduce stress, control cholesterol levels
→ works as an immune booster, increase fertility.

⑤ Amla (Indian gooseberry) - Sour taste.

→ rich source of Vitamin C

→ amla acts as an antioxidant that promotes the function of the immune system.

→ Carotene component in amla improve eyesight.

→ Amla have also anti-inflammatory properties that can help to ~~with~~ arthritis-related joint and help to treat wounds.

→ Commercially - hair oil - reduce hair fall.

⑥ Turmeric - rich in phytonutrients - protect the body.
- prevention of medical conditions such as Cancer and heart disease.

⑦ Neem - neem have many medicinal properties in leaves such as -

(i) anti-inflammatory - improve wound, Infection

(ii) anti-malarial

(iii) anti-bacterial

(iv) anti-viral

(v) anti-oxidant

(vi) anti-carcinogenic

(vii) anti-ulcer etc.

⑧ Ginger - stem contain beneficial compound used to treat colds, nausea, migraines and blood pressure.

⑨ Garlic - control Cholesterol, regulate blood pressure.
- control cough.

Scope of Entrepreneurship

Date _____ Page _____

- In India agriculture is chief mean of livelihood.
- Around 58% of the land cultivate in india compare to 11% of the world.
- There is need for a new approach and technology for the another green revolution.
- Majority of the population is depended on agricultures still the the equivalent number of unemployment exists.
- Entrepreneurship helps in developing nation's economy, improve production and labour market, builds job opportunities, and finally increase employment rate.
- Entrepreneurial development - an organised and regulated growth of Individual.
 - Such as - Knowledge, financial support, marketing etc

➤ Scope of Agri-Entrepreneurship.

④ India is gifted with diverse agro-climate, which assist production

- ① Honey - agribusiness -
- ② Mushroom farming
- ③ food processing
- ④ Cultivation of Medicinal Plant
- ⑤ Cultivation of Herbs and Spices
- ⑥ Cultivation of fruits
- ⑦ Dairy Product
- ⑧ Fisheries Development.
- ⑨ Poultry farming
- ⑩ Vegetable farming
- ⑪ flowers farming
- ⑫ Organic Production.

• Introduction of Medicinal Chemistry

Date _____

Page _____

- Medicinal Chemistry is the Science, which deals with the discovery and design of new and better ~~drug~~ or therapeutic chemicals.
- then these chemicals into new medicines and Drugs.

• Generally Medicinal Chemists can -

- Make new compounds.
- Determine their effect on biological process
- Alter the structure of the compound for optimum effect and minimum side effects.
- Study uptake, distribution, metabolism and excretion of drugs.

• Primary Objective of Medicinal Chemistry.

(i) Drug discovery - finding a lead. - Steps given -

- ① • Choose a disease
- ② • Choose a drug target.
- ③ • Identify a bioassay.
- ④ • Find a lead compound
- ⑤ • Isolate and purify the lead compound if necessary.
- ⑥ • Determine the structure of the lead compound if necessary.

(ii) Drug Design

- ① Identify Structure-activity relationships (SARs)
- ② Identify the pharmacophore.
- ③ Improve target Interaction
- ④ Improve pharmacokinetic properties.

(iii) Drug Development

Date _____ Page _____

- ① Patent the drug
- ② Carry out preclinical trials
- ③ Design a manufacturing process
- ④ then again carry out Clinical trials
- ⑤ Register and market the drug
- ⑥ Make Money.

• Importance of Medicinal Chemistry in Ayurveda Research

- (i) Improvement in research methodology - is a way to systematic collection, analysis and Interpretation of data to solve a research problem.
- (ii) Encouragement of Research on Ayurvedic fundamentals.
- (iii) Validation of Ayurvedic drugs with Reverse pharmacology
- (iv) Evidence-based benefits of traditional approaches.
- (v) Encouragement of Interdisciplinary Research.
- (vi) Improvement in quality of Herbal drugs.
- (vii) Active Involvement of government
- (viii) Development of Ayurvedic pharma Sector.
- (ix) Upgradation of Ayurvedic literature.
- (x) Revalidation of Clinical trials.
- (xi) Combination of Ayurveda and Western medicine.
- (xii) Improvement in personalized medicine.
- (xiii) Resolution of Controversies in Ayurvedic drugs.

Medicinal Chemistry is tool of encouraging

Date

Page

• Yes It is true that Medicinal Chemistry is tool of encouraging for following reason -

- (i) Due to Medicinal Chemistry - Ayurveda more developed.
- (ii) Medicinal Chemistry provide evidence for Ayurveda So more individual believe.
- (iii) Medicinal Chemistry detect Chemicals (which help ~~in~~ to cure illness) in Medicinal Plant then these therapeutic Chemical Synthesis in labs. - So ~~it~~ it ~~led~~ further save the Medicinal plant.

↳ following this we can say Medicinal Chemistry is tool of encouraging in Ayurveda.

→ Notes devel download by - Telegram Channel -

~~BA~~ Content for BA BML

→ Credit by - your gods - so say to thanks own god.